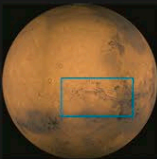


VALLES MARINERIS

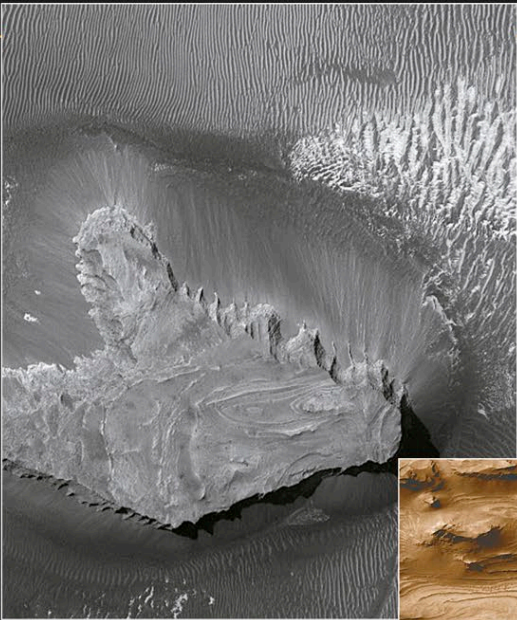
Valles Marineris



|        |                             |
|--------|-----------------------------|
| TYPE   | Canyon system               |
| AGE    | About 3.5 billion years     |
| LENGTH | Over 2,500 miles (4,000 km) |

Valles Marineris is the largest feature formed by tectonic activity on Mars. It consists of a system of canyons that stretches for over 2,500 miles (4,000 km), is up to 430 miles (700 km) wide, and averages 5 miles (8 km) in depth. The Grand Canyon in Arizona is dwarfed in comparison; it is only about one-tenth as long and one-fifth as deep. Valles Marineris lies just south of the Martian equator, and the system trends, very roughly, west to east. The trend follows a set of fractures that radiates from the Tharsis Bulge at Marineris’s western end.

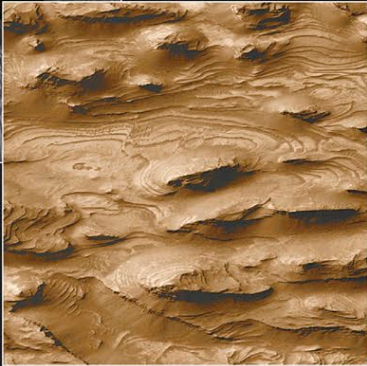
The origins of the system date back a few billion years to when the canyons were formed by faulting. This contrasts with the Grand



Canyon, which is a primarily water-eroded canyon. But water, as well as wind, has played its part in the development of the Marineris system. Buffeting winds, flowing water, and the collapse of unstable walls have all widened and deepened the canyons.

MESA

This small mesa (a flat-topped hill) lies in northwestern Candor Chasma in the central Valles Marineris. Light-toned outcrops of layered sedimentary rock are exposed on the top. These may have formed from material deposited in a lake in the chasma. Darker windblown ripples cover the surrounding plains.



The Noctis Labyrinthus region marks the western end of the system. This is a roughly triangular area of intersecting rift valleys that form a mazelike arrangement. The eastern end of Valles Marineris is bounded by chaotic terrain of irregular appearance. Here, smaller canyons and depressions give way to outflow canyons. These carried ancient rivers of water out of Marineris toward the lowland region, Chryse Planitia, to the north. This whole area has seen extensive water erosion; millions of cubic miles of material have been removed by water action.

The system’s canyons are described as chasma (plural chasmata) and are given identifying names. The main chasma in the western part of the system is Ius. The central complex is made up of three parallel canyons:

LAYERED DEPOSITS

A detail of the floor of western Candor Chasma shows layered sedimentary rock. Up to 100 layers have been counted, each about 30 ft (10 m) thick. The layers may be made from material deposited in an impact crater before the chasma formed.

SCARRED PLANET

Valles Marineris extends about a quarter of the way around Mars. All but the extreme western end of the canyon system is shown here. This image covers the area from the equator (top) to 20° south.